MILITARY STANDARD

TRUCKLOADING OF HAZARDOUS MATERIALS

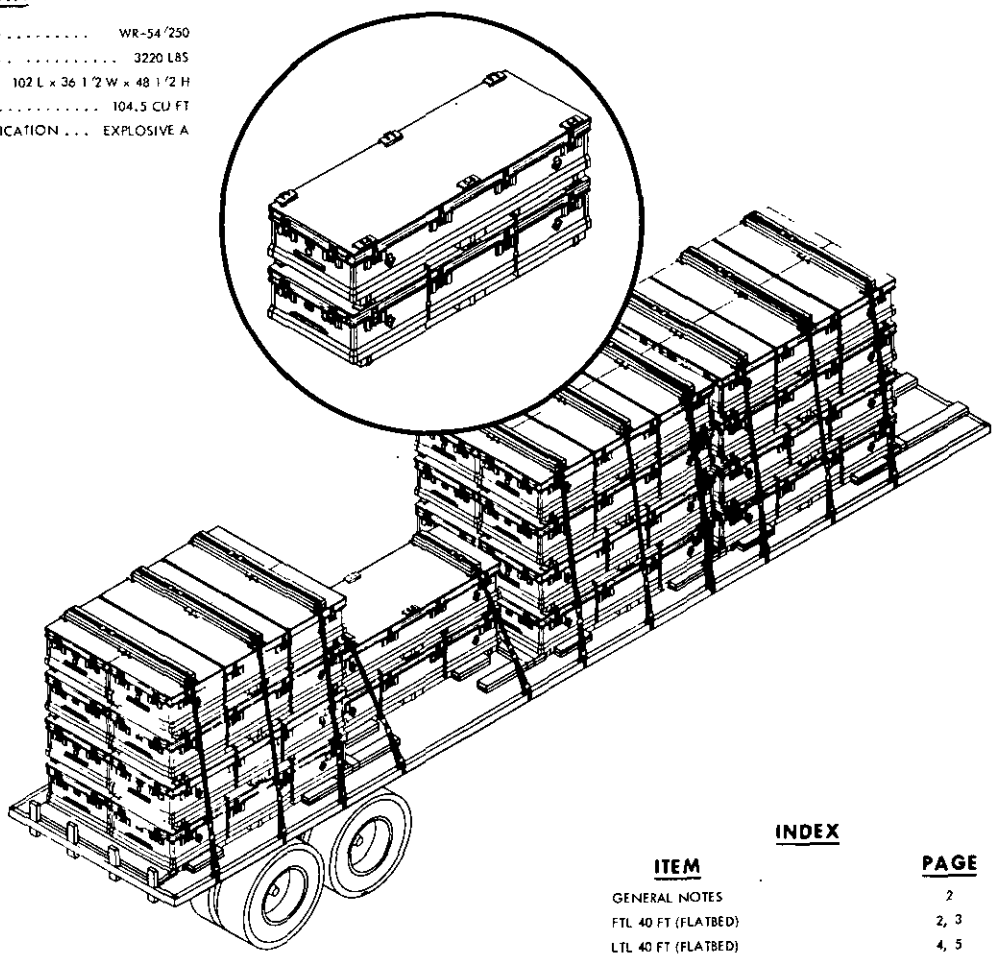
BOMB, CLUSTER, MK 20 AND MODS (ROCKEYE II)
IN CONTAINER CNU-208/E
FLEET ISSUE UNIT LOAD

MIL-STD-1320-158 (NAVY)

3 SEPTEMBER 1974

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/250
WEIGHT . . . . . 3220 LBS
DIMENSIONS . . . . . 102 L x 36 1/2 W x 48 1/2 H
CUBE . . . . . 104.5 CU FT
DOT HAZARD CLASSIFICATION . . . EXPLOSIVE A



NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

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FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M.S. Bruno NWHL 8/19/74
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 8/19/74
SIGNATURE AIRSYSCOM BY DIRECTION DATE

ORIGINATOR Charles McBride 8/19/74
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

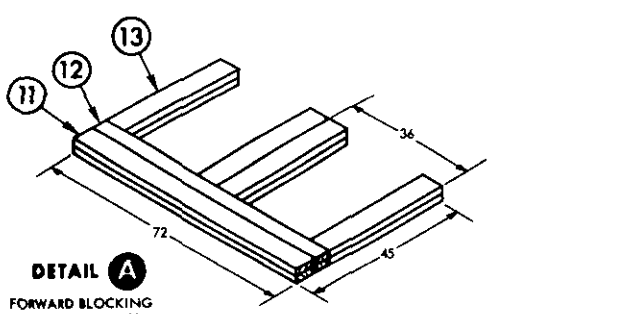
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GENERAL NOTES

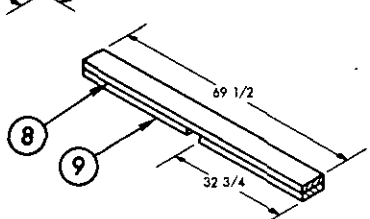
1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING FLEET ISSUE UNIT LOADS OF CLUSTER BOMBS, MK 20 AUD MODS (ROCKEYE II) IN CONTAINER CNU-208/E AND PALLETIZED IN ACCORDANCE WITH *WR-54/250.*
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS OR 40 FT VAN TRAILERS. THE TRAILER AXLES SHALL BE LOCATED IN THE WESTERN POSITION (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.
3. WHEN REQUIRED TO SHIP UNIT LOADS TWO HIGH, THEY SHALL BE STACKED AND STRAPPED TOGETHER AS SPECIFIED ON PAGE 9.
4. EACH STACK OF CONTAINERS SHALL BE SECURED TOGETHER WITH TWO 1 1/4 x .035 CROSS STRAPS. THE CROSS STRAP SHALL PASS OVER THE COVERS OF THE CONTAINERS IN THE TOP LAYER AND RETURN PASSING THROUGH THE FORK POCKETS OF THESE CONTAINERS, BINDING THE TOP LAYER TOGETHER.
5. CHAINS AND STEEL STRAPS MAY BE USED INTERCHANGEABLY FOR TIE-DOWNS ON A 1 TO 1 BASIS.
6. STEEL STRAPS SHALL BE 2 x .050 AND APPLIED AS SPECIFIED BY THE LOAD STRAPPING REQUIREMENTS SHOWN ON PAGE 9.
7. WHEN USING CHAINS OR STEEL STRAPS FOR TIE-DOWNS, A PROTECTOR BOARD ASSEMBLY, DETAIL B, SHALL BE INSERTED BETWEEN THE CONTAINER AND THE TIE-DOWN, AS SHOWN ON PAGE 3. CHAINS SHALL BE SECURED TO THE PROTECTOR BOARDS WITH FOUR 20d NAILS DRIVEN THROUGH THE CHAIN LINK INTO THE BOARD AND CLINCHING THE NAIL OVER THE LINK.
8. CHAINS AND LOAD BINDERS SHALL MEET THE FOLLOWING CONDITIONS:
   A. CHAINS AND FITTINGS SHALL BE 3/8 INCH AND LOAD BINDERS 3/8 INCH CAPACITY.
   B. CHAINS, FITTINGS AND LOAD BINDERS SHALL HAVE A MINIMUM SAFE WORKING LOAD OF 5000 LBS.
   C. LOAD BINDERS SHALL BE SAFETY WIRED WITH 16 GAUGE SOFT ANNEALED IRON WIRE OR SECURED USING THE END PIECE OF THE 3/8 CHAIN.
9. PRIOR TO LOADING THE TRAILER AND DURING PRELOADING INSPECTION REQUIRED BY NAVWEPS OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER *NOTICEABLE DEFECTS.* THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE ON ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.
10. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO DUNNAGE IN AN ACCESSIBLE AREA AND ATTACH PLACARD "EXPLOSIVES A" TO BOTH SIDES, FRONT AND BACK OF TRAILER.
12. APPLICABLE MATERIAL SPECIFICATIONS:
    DUNNAGE LUMBER, MM-L-751
    NAILS, FF-N-105, TYPE II, STYLE 10, COMMON BRIGHT
    STRAPPING, QQ-S-781, TYPE I, CLASS A.
13. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL 40 FT TRAILER (FLATBED)

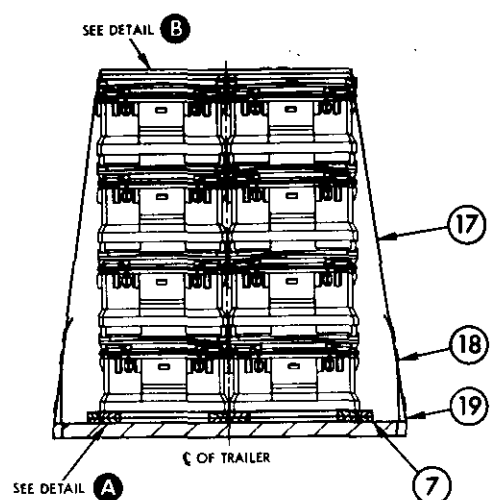
1. START LOADING SEQUENCE AT REAR OF TRAILER. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.



DETAIL A
FORWARD BLOCKING ASSEMBLE IN PLACE (1 REQUIRED)

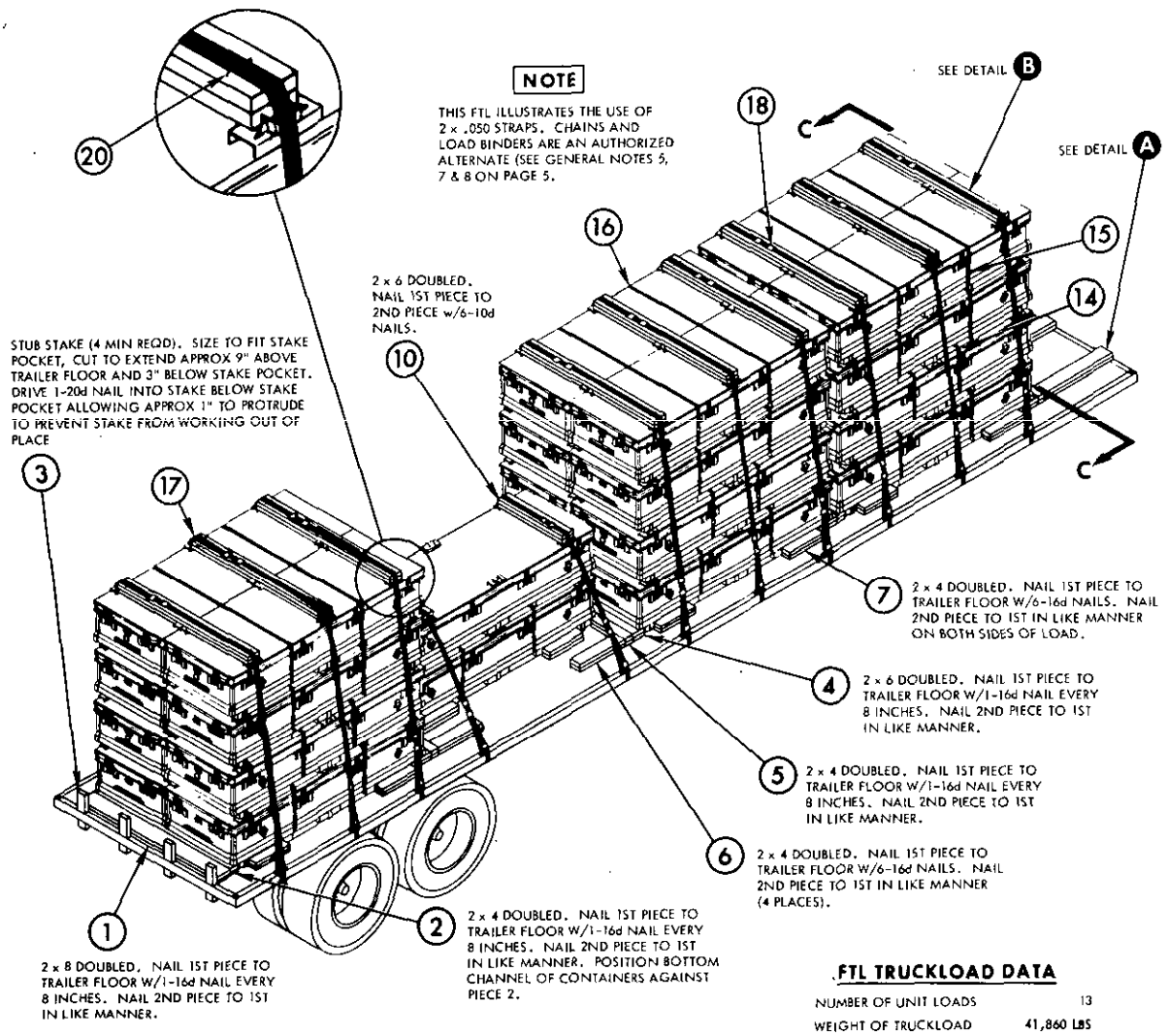


DETAIL B
PROTECTOR BOARD



SECTION C-C

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FTL TRUCKLOAD DATA

NUMBER OF UNIT LOADS	13
WEIGHT OF TRUCKLOAD	41,860 LBS

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
20	STAPLES	FOR 2 STRAP	AS REQD	-	-	-
19	STAKE POCKET PAD (STRAPPING)	2 x .050 x 24	22	-	-	-
18	SEAL	FOR 2 STRAP	88	-	-	-
17	STRAP, TIE DOWN	2 x .050 x 29 FT	11	-	-	-
16	STRAP, CROSS	1 1/4 x .035 x 17 FT 6	6	-	-	-
15	SEAL	FOR 1 1/4 STRAP	18	-	-	-
14	STRAP, STACKING	1 1/4 x .035 x 11 FT	12	-	-	-
13	FORWARD BACKUP CLEAT	2 x 6 x 36	8	SEE DETAIL A		
12	FORWARD CROSSMEMBER	2 x 4 x 72	2	SEE DETAIL A		
11	FORWARD CROSSMEMBER	2 x 6 x 72	2	SEE DETAIL A		

LIST OF MATERIALS AND NAILING DATA

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
10	PROTECTOR BOARD	2 x 6 x 34	4	SEE FIELD NOTE		
9	PROTECTOR BOARD	2 x 6 x 32 3/4	18	B	5	10d
8	PROTECTOR BOARD	2 x 6 x 69 1/2	9	SEE 9	-	-
7	SLEEPER	2 x 4 x 18	32	SEE FIELD NOTE		
6	BACKUP CLEAT	2 x 4 x 24	8	SEE FIELD NOTE		
5	INTERMEDIATE CROSSMEMBER	2 x 4 x 72	6	SEE FIELD NOTE		
4	INTERMEDIATE CROSSMEMBER	2 x 6 x 72	6	SEE FIELD NOTE		
3	STUB STAKE	TO SUIT	4	SEE FIELD NOTE		
2	REAR CROSSMEMBER	2 x 4 x 72	2	SEE FIELD NOTE		
1	REAR CROSSMEMBER	2 x 8 x 72	2	SEE FIELD NOTE		

LIST OF MATERIALS AND NAILING DATA

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LTL 40 FT TRAILER (FLATBED)

1. WHEN REQUIRED TO SHIP 9, 10, 11 OR 12 UNIT LOADS, SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF UNIT LOADS TO BE SHIPPED FROM TABLE I AND POSITION THEM ON THE TRAILER ACCORDINGLY.

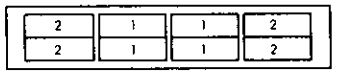
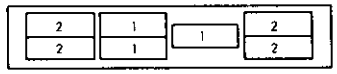
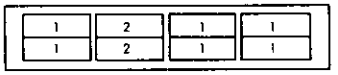
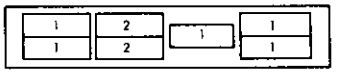
NOTE

WHEN REQUIRED TO SHIP 8 OR LESS UNIT LOADS, THEY MAY BE LOADED ON A FLATBED TRAILER, DUNNAGE AND TIE DOWN USING THE PRINCIPLES SHOWN ON PAGE 5 FOR THE 1 HIGH UNIT LOADS.

NOTE

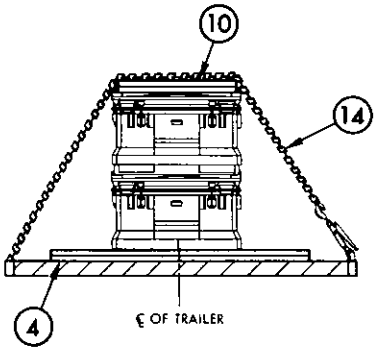
8 OR LESS UNIT LOADS MAY ALSO BE SHIPPED IN A VAN TRAILER. SEE PAGES 7 AND 8.

TABLE I

NO. OF UNIT LOADS	LOAD PATTERN*	FWD
12		
11		
10		
9		

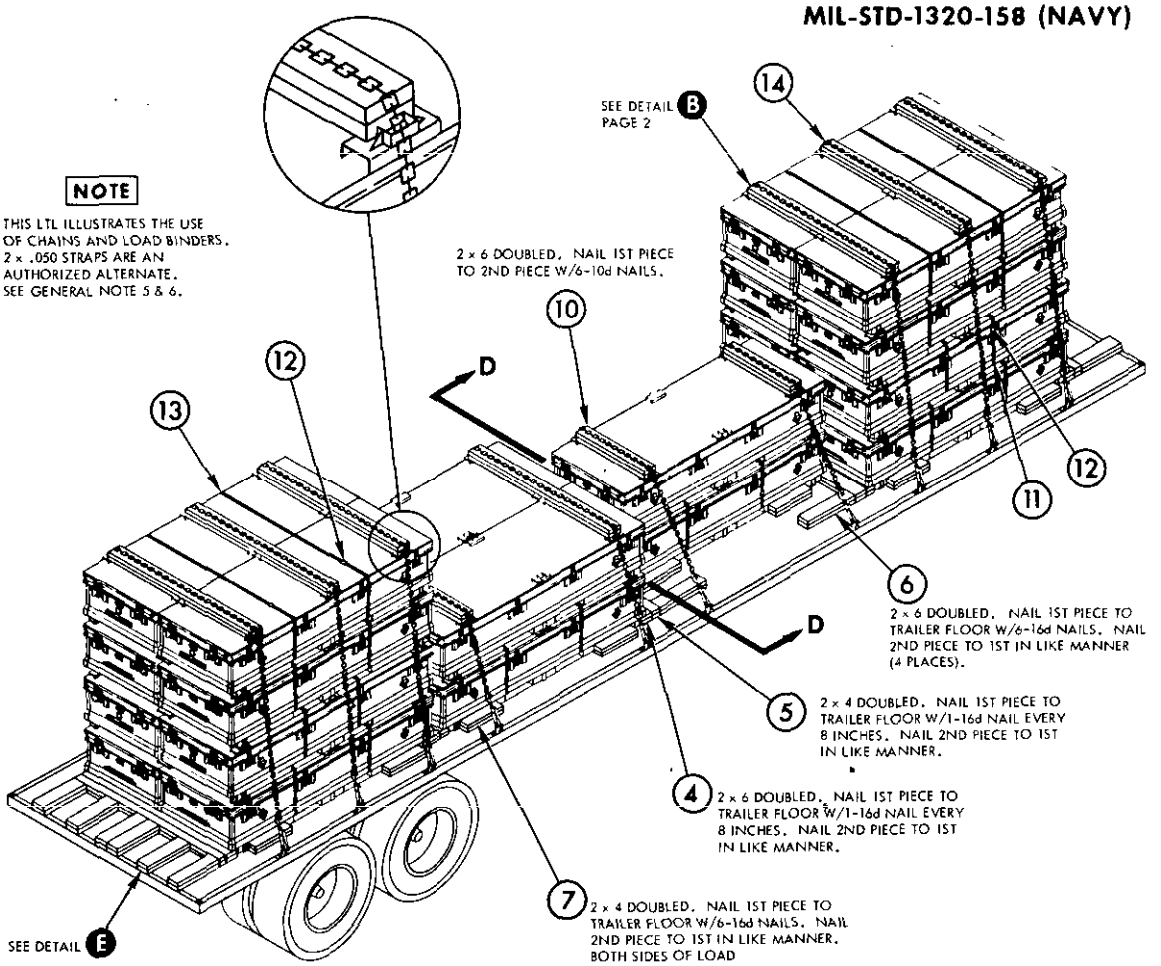
*LOAD PATTERN SHOWS THE FLOOR PLAN FOR THE LOADED TRAILER. NUMBER IN LOAD PATTERN BLOCKS INDICATES THE NUMBER OF LAYERS OF UNIT LOADS OF CONTAINERS.

2. THE LTL ON PAGE 5 IS FOR 11 UNIT LOADS. LTL SHIPMENTS ARE NOT LIMITED TO THIS QUANTITY. ACTUAL SHIPPING REQUIREMENTS WILL DETERMINE THE NUMBER OF UNIT LOADS TO BE SHIPPED AND THEY SHOULD BE DUNNAGED AND TIED DOWN IN ACCORDANCE WITH APPROPRIATE PARTS OF THIS STANDARD.



SECTION D-D

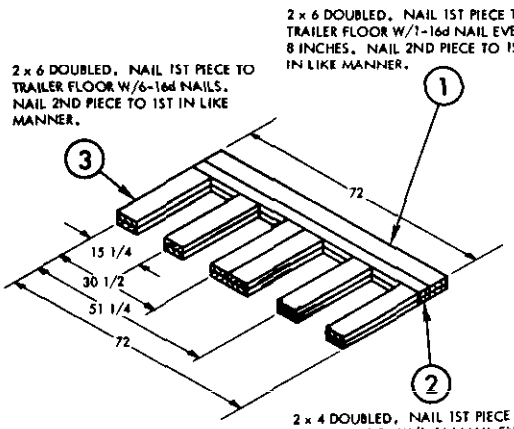
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TRUCKLOAD DATA

NUMBER OF UNIT LOADS 11

WEIGHT OF TRUCKLOAD. . . 35,420 LBS



DETAIL E

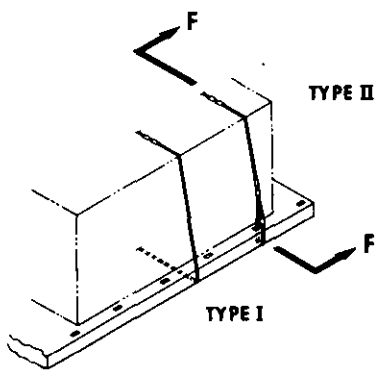
FRONT AND REAR BLOCKING ASSEMBLE IN PLACE

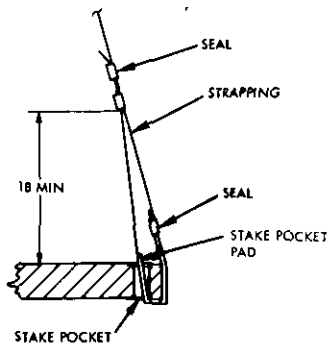
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
14	CHAIN & LOADBINDER	3/8	10	-	-	-
13	STRAP, CROSS	1 1/4 x .035 x 17 FT 6	4	-	-	-
12	SEAL	FOR 1 1/4 STRAP	12	-	-	-
11	STRAP, STACKING	1 1/4 x .035 x 11 FT	8	-	-	-
10	PROTECTOR BOARD	2 x 6 x 34	4	SEE FIELD NOTE		
9	PROTECTOR BOARD	2 x 6 x 32 3/4	16	SEE DETAIL B, PAGE 2		
8	PROTECTOR BOARD	2 x 6 x 69 1/2	8	SEE DETAIL B, PAGE 2		
7	SLEEPER	2 x 4 x 24	32	SEE FIELD NOTE		
6	BACKUP CLEAT	2 x 6 x 23	8	SEE FIELD NOTE		
5	INTERMEDIATE CROSSMEMBER	2 x 4 x 72	6	SEE FIELD NOTE		
4	INTERMEDIATE CROSSMEMBER	2 x 6 x 72	6	SEE FIELD NOTE		
3	BACKUP CLEAT	2 x 6 x 23	24	SEE DETAIL E		
2	CROSSMEMBER	2 x 4 x 72	4	SEE DETAIL E		
1	CROSSMEMBER	2 x 6 x 72	4	SEE DETAIL E		

LIST OF MATERIALS AND NAILING DATA

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OVER THE LOAD STRAPPING





SECTION F-F

TIE-DOWN STRAPPING

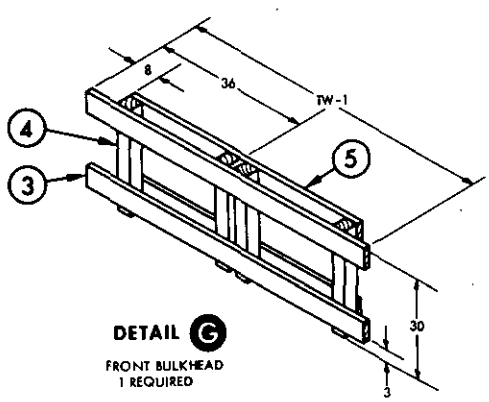
THE FOLLOWING TWO METHODS OF TIE-DOWN STRAPPING ARE APPROVED FOR USE WITH THE SHIPMENT OF LOADS ON FLATBED TRAILERS. STEEL STRAPPING SHALL BE IN ACCORDANCE WITH FEDERAL SPECIFICATION QQ-S-781. STRAPPING SEALS SHALL HAVE A MINIMUM TENSILE STRENGTH EQUAL TO 75 PER CENT OF THE STRAP STRENGTH. FOR EACH OF THE TYPES ILLUSTRATED ABOVE IT IS PREFERRED TO POSITION, TENSION AND DOUBLE CRIMP THE STRAP SEALS AT THE TOP OF THE LOAD, IF PRACTICABLE.

TYPE I - CONTINUOUS STRAPPING AROUND THE FLATBED AND THE LOAD.

TYPE II - THE STRAPPING IS SECURED TO THE STAKE POCKETS, ONE PIECE ON EACH SIDE OF THE TRAILER, AND IS BROUGHT UP OVER THE LOAD, TENSIONED, AND SEALED WITH TWO DOUBLE-CRIMPED SEALS ON THE TOP. METHOD OF SECURING STRAPPING TO STAKE POCKETS IS SHOWN IN SECTION F-F THE SHORT END IS ON THE INSIDE AND IS SECURED WITH TWO DOUBLE-CRIMPED SEALS AT A MINIMUM OF 18 INCHES ABOVE THE TRAILER BED. A STAKE POCKET PAD (A SHORT-PIECE OF THE SAME STRAPPING 18 INCHES LONG) IS INSERTED BETWEEN THE MAIN STRAP AND HELD IN PLACE WITH A SEAL AS SHOWN.

FTL AND LTL 40 FT TRAILER (VAN)

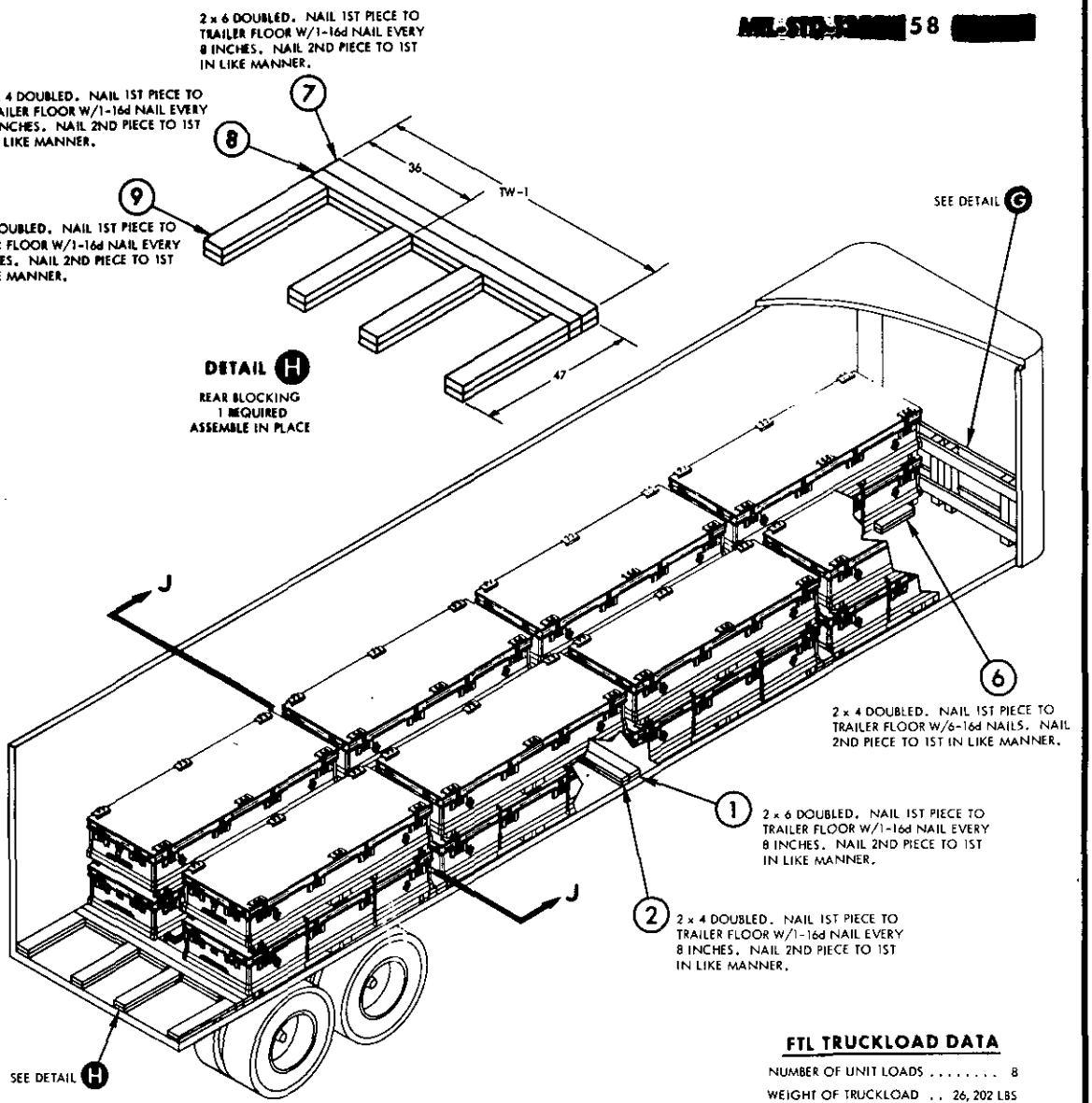
1. A UNIT LOAD OF CLUSTER BOMBS IN CNU-208/E CONTAINERS IS LOADED INTO A VAN USING A 6000 LB CAPACITY FORKLIFT TRUCK, HANDLING THE UNIT LOAD FROM THE END.
2. A FTL VAN CONSISTS OF 8 UNIT LOADS AND IS LOADED, BLOCKED AND BRACED AS SHOWN ON PAGE 7. WHEN REQUIRED TO SHIP LESS THAN 8 UNIT LOADS IN A VAN, LOAD, BLOCK AND BRACE FOLLOWING THE PRINCIPLES OF THE FTL VAN ON PAGE 7 AND THE LTL VAN ON PAGE 8.



DETAIL G

FRONT BULKHEAD
1 REQUIRED

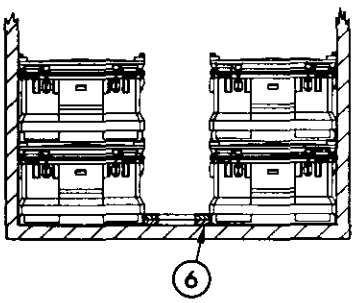
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FTL TRUCKLOAD DATA

NUMBER OF UNIT LOADS 8

WEIGHT OF TRUCKLOAD . . 26,202 LBS

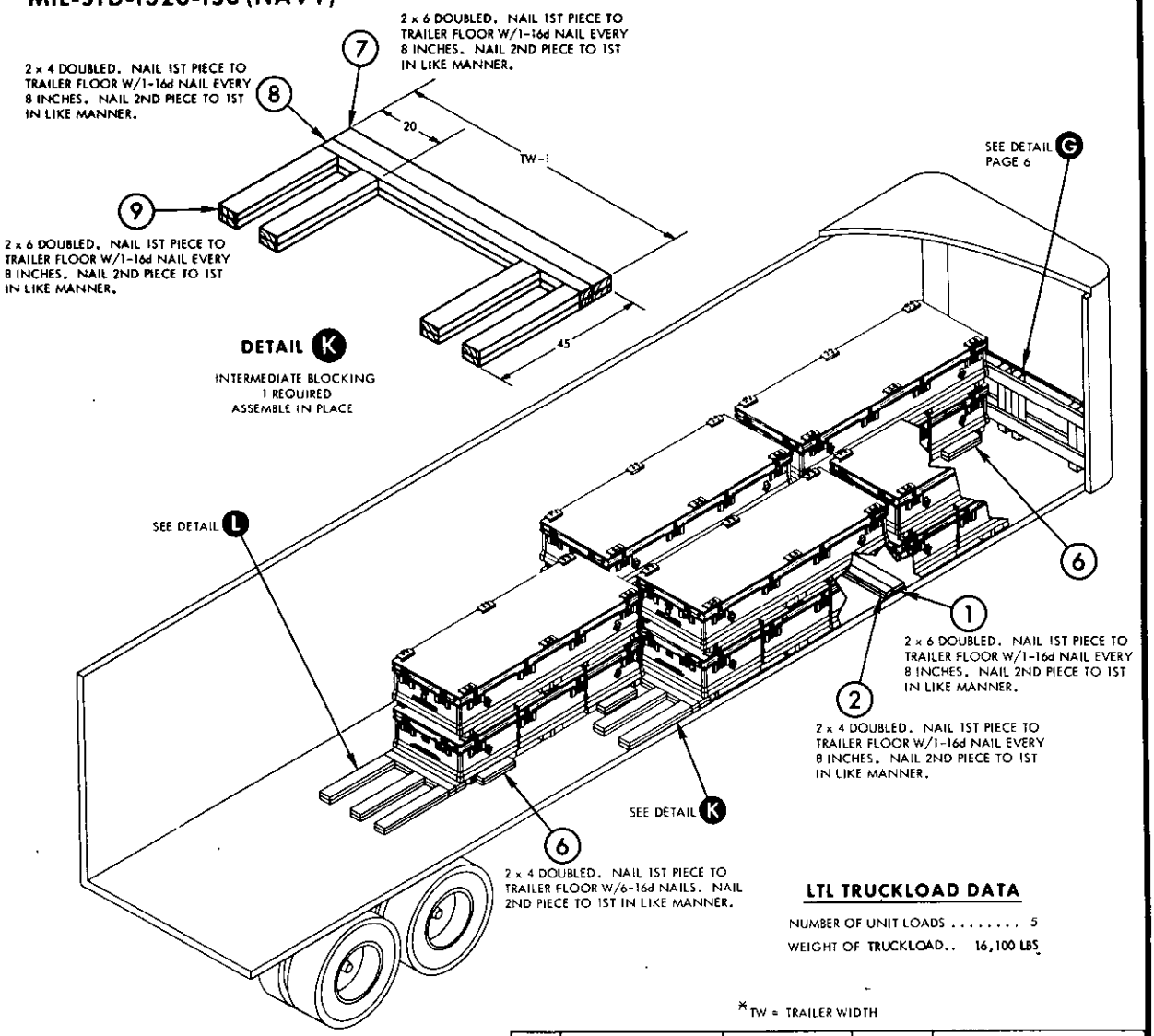


SECTION J-J

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER	SIZE
	LIST OF MATERIALS AND NAILING DATA				NAILS	
9	BACKUP CLEAT	2 x 6 x 38	8	SEE DETAIL H		
8	CROSSMEMBER	2 x 4 x TW*-1	2	SEE DETAIL H		
7	CROSSMEMBER	2 x 6 x TW*-1	2	SEE DETAIL H		
6	SLEEPER	2 x 4 x 24	32	SEE FIELD NOTE		
5	FWD CROSSMEMBER	2 x 6 x TO SUIT	2	4	3/JOINT	12d
4	VERTICAL	4 x 4 x 30	4	SEE 3 & 5	-	-
3	AFT CROSSMEMBER	2 x 6 x TW*-1	2	4	3/JOINT	12d
2	SPACER CROSSMEMBER	2 x 4 x TW*-1	6	SEE FIELD NOTE		
1	SPACER CROSSMEMBER	2 x 6 x TW*-1	6	SEE FIELD NOTE		

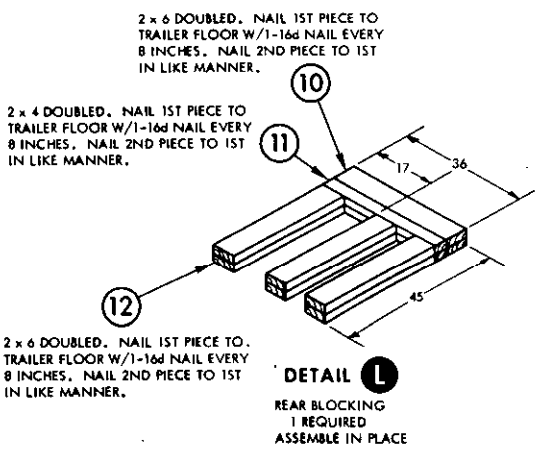
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LTL TRUCKLOAD DATA

NUMBER OF UNIT LOADS 5

WEIGHT OF TRUCKLOAD.. 16,100 LBS



*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAIL NUMBER	NAIL SIZE
					NAILS	
12	BACKUP CLEAT	2 x 6 x 36	6	SEE DETAIL L		
11	REAR CROSSMEMBER	2 x 4 x 36	2	SEE DETAIL L		
10	REAR CROSSMEMBER	2 x 6 x 36	2	SEE DETAIL L		
9	BACKUP CLEAT	2 x 6 x 36	8	SEE DETAIL K		
8	INTERMEDIATE CROSSMEMBER	2 x 4 x TW*-1	2	SEE DETAIL K		
7	INTERMEDIATE CROSSMEMBER	2 x 6 x TW*-1	2	SEE DETAIL K		
6	SLEEPER	2 x 4 x 24	24	SEE FIELD NOTE		
5	FWD CROSSMEMBER	2 x 6 x TO SUIT	2	4	3/JOINT	12d
4	VERTICAL	4 x 4 x 30	4	SEE 3 & 5	-	-
3	AFT CROSSMEMBER	2 x 6 x TW*-1	2	4	3/JOINT	12d
2	SPACER CROSSMEMBER	2 x 4 x TW*-1	2	SEE FIELD NOTE		
1	SPACER CROSSMEMBER	2 x 6 x TW*-1	2	SEE FIELD NOTE		

LIST OF MATERIALS AND NAILING DATA

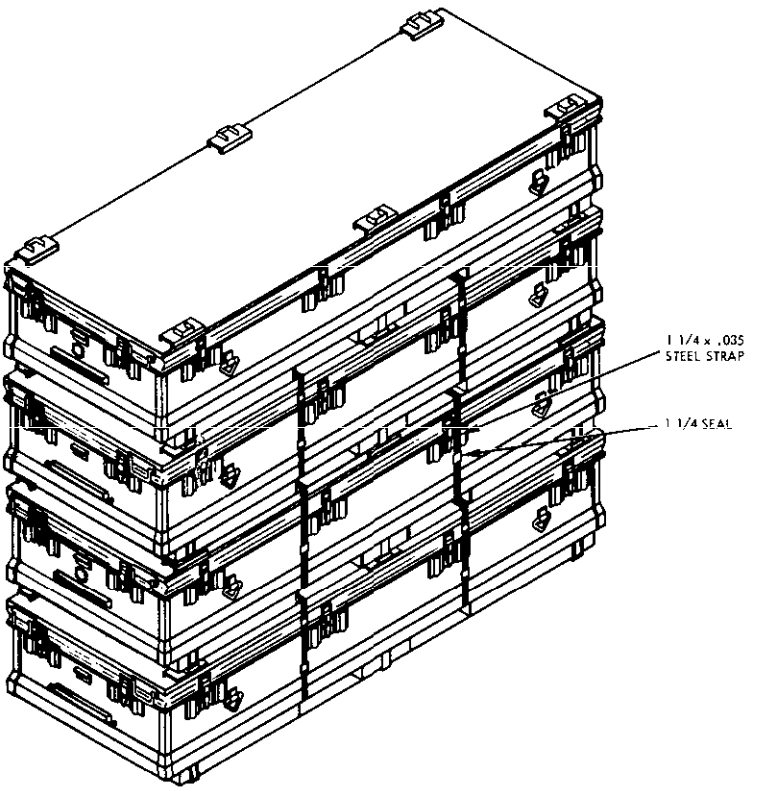
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STRAPPING AND STACKING PROCEDURE

1. STACK THE UNIT LOADS TOGETHER USING CARE TO ALIGN THE INTERLOCKING FEATURES OF THE CONTAINERS.
2. THREAD THE STRAPPING THROUGH THE FORK POCKETS OF THE TOP CONTAINER OF THE BOTTOM UNIT LOAD AND THE BOTTOM CONTAINER OF THE TOP UNIT LOAD. TENSION AND SECURE WITH ONE DOUBLE NOTCHED SEALS.
3. STACKED AND STRAPPED UNIT LOADS OF CLUSTER BOMBS IN CNU-208/E CONTAINERS ARE HANDLED AND POSITIONED ON FLATBED TRAILERS USING FORK-LIFT TRUCKS OF SUITABLE CAPACITY. IF ADEQUATE CAPACITY EQUIPMENT IS NOT AVAILABLE, THE UNIT LOADS SHOULD BE POSITIONED ON THE TRAILER ONE AT A TIME AND THE STACKED UNIT LOADS STRAPPED TOGETHER IN POSITION.

WARNING

TO AVOID TOPPLING, SPECIAL CARE SHOULD BE TAKEN WHEN MOVING STACKS OF CONTAINERS. THE STRAPPING TOGETHER OF UNIT LOADS OF CONTAINERS IS AUTHORIZED AS PART OF THE TRUCKLOADING PROCEDURE ONLY. STRAPPING TOGETHER OF UNIT LOADS IS TO ENSURE CONTINUOUS ENGAGEMENT OF THE STACKING FEATURES AND MOVEMENT OF STRAPPED UNIT LOADS SHOULD BE LIMITED TO THAT NECESSARY TO LOAD OR UNLOAD THE FLATBED TRAILER.



REVIEW ACTIVITY:
NAVY-OS, AS

PREPARING ACTIVITY:
NAVY-OS
(PROJECT NO. 8140-N244

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